30-Year Analysis of Housing Price Trends in the United States

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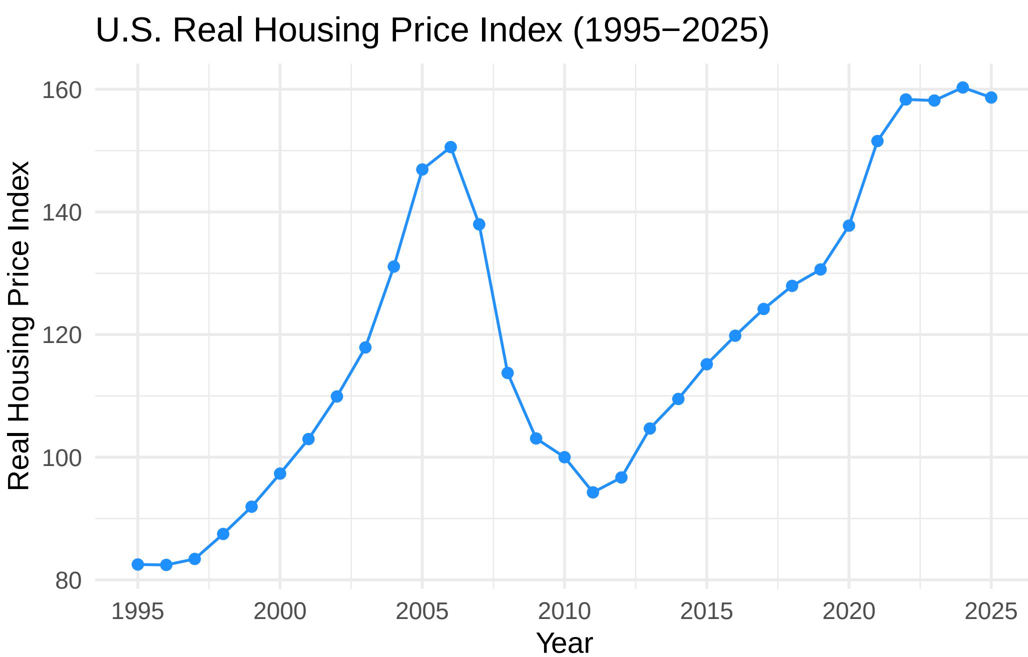
U.S. Real Housing Price Index

Code for Price Index

```
[1] "year"                "Real_Housing_Index"
```




```
   year Real_Housing_Index
26 1995           82.50590
27 1996           82.43685
```

```
28 1997            83.41053
29 1998            87.48183
30 1999            91.92583
31 2000            97.32615
32 2001           102.95905
33 2002           109.90865
34 2003           117.90240
35 2004           131.08735
36 2005           146.91815
37 2006           150.57413
38 2007           137.97272
39 2008           113.73963
40 2009           103.05962
41 2010           100.00397
42 2011            94.26445
43 2012            96.69220
44 2013           104.67750
45 2014           109.49972
46 2015           115.15705
47 2016           119.81035
48 2017           124.17410
49 2018           127.93960
50 2019           130.61593
51 2020           137.76130
52 2021           151.55992
53 2022           158.33925
54 2023           158.16620
55 2024           160.29122
56 2025           158.65657
```

```
Loading required package: knitr
```

Table 1: Average Annual Real Housing Price Index (1995–2025)

	Year	Real Housing Price Index
26	1995	82.5
27	1996	82.4
28	1997	83.4
29	1998	87.5
30	1999	91.9
31	2000	97.3

	Year	Real Housing Price Index
32	2001	103.0
33	2002	109.9
34	2003	117.9
35	2004	131.1
36	2005	146.9
37	2006	150.6
38	2007	138.0
39	2008	113.7
40	2009	103.1
41	2010	100.0
42	2011	94.3
43	2012	96.7
44	2013	104.7
45	2014	109.5
46	2015	115.2
47	2016	119.8
48	2017	124.2
49	2018	127.9
50	2019	130.6
51	2020	137.8
52	2021	151.6
53	2022	158.3
54	2023	158.2
55	2024	160.3
56	2025	158.7

Narrative Text

Introduction

Housing prices are an important indicator of economic conditions in the United States. Changes in housing prices can reflect periods of economic growth, financial instability, and recovery. This analysis focuses on the U.S. Real Housing Price Index from 1995 to 2025 to better understand long-term trends and major shifts in the housing market.

Data

The dataset used in this analysis comes from the Federal Reserve Economic Data (FRED) database, specifically the Real Housing Price Index (QUSR628BIS). This dataset provides monthly values that measure changes in housing prices adjusted for inflation. Using real values allows for a more accurate comparison over time because it removes the effects of inflation.

Data Preparation

To prepare the data for analysis, the monthly housing price values were grouped by year and averaged to create a single value per year. The year was extracted from the date variable and converted into a numeric format for easier plotting. The dataset was then filtered to include only the years from 1995 through 2025. Finally, the housing price index values were rounded to one decimal place to improve readability in the data table.

Visualization

The visualization is a line graph that shows the trend of the Real Housing Price Index over time. The graph reveals a steady increase in housing prices from the late 1990s through the mid-2000s, followed by a sharp decline during the 2008 financial crisis. After reaching a low point around 2011, housing prices gradually recovered and continued to rise, reaching their highest levels in the early 2020s. This pattern highlights the impact of major economic events on the housing market.

Data Table

The data table presents the average annual Real Housing Price Index values from 1995 to 2025. By summarizing the data into yearly averages, the table provides a clear and organized view of how housing prices changed over time. It allows for easy comparison between years and supports the trends observed in the visualization.

Conclusion

Overall, the Real Housing Price Index shows significant growth over the past three decades, with a major disruption during the 2008 financial crisis. The recovery period that followed demonstrates the resilience of the housing market, as prices eventually surpassed previous highs. This analysis emphasizes how economic events can strongly influence housing prices and highlights the importance of examining long-term trends when evaluating the housing market.

U.S. Inflation Rates

Code for Inflation

```
Attaching package: 'dplyr'


The following objects are masked from 'package:stats':

    filter, lag


The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union


[1] "observation_date" "FPCPITOTLZGUSA"


# A tibble: 6 x 2
  observation_date FPCPITOTLZGUSA
  <date>                    <dbl>
1 1960-01-01                 1.46
2 1961-01-01                 1.07
3 1962-01-01                 1.20
4 1963-01-01                 1.24
5 1964-01-01                 1.28
6 1965-01-01                 1.59


# A tibble: 6 x 2
   year Inflation
  <dbl>     <dbl>
1  1995      2.81
2  1996      2.93
3  1997      2.34
4  1998      1.55
5  1999      2.19
6  2000      3.38


Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
i Please use `linewidth` instead.
```

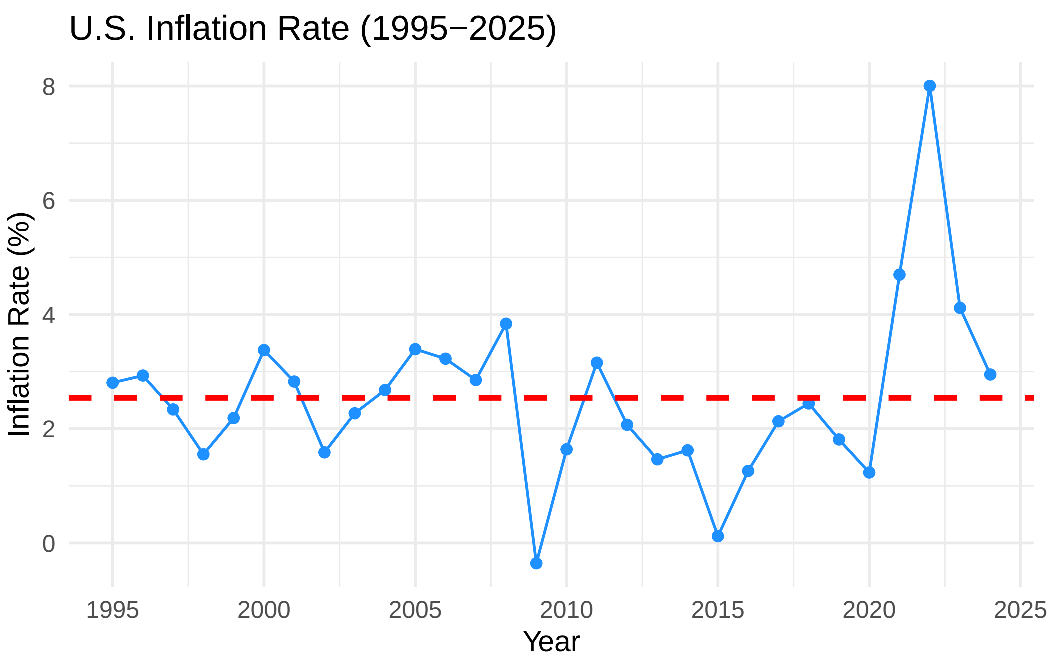


Table 2: Annual U.S. Inflation Rate (1995–2025)

Year	Inflation Rate (%)
1995	2.81
1996	2.93
1997	2.34
1998	1.55
1999	2.19
2000	3.38
2001	2.83
2002	1.59
2003	2.27
2004	2.68
2005	3.39
2006	3.23
2007	2.85
2008	3.84
2009	-0.36
2010	1.64
2011	3.16
2012	2.07
2013	1.46
2014	1.62

Year	Inflation Rate (%)
2015	0.12
2016	1.26
2017	2.13
2018	2.44
2019	1.81
2020	1.23
2021	4.70
2022	8.00
2023	4.12
2024	2.95

Narrative Text

Introduction

Inflation measures how much prices for goods and services increase over time.It is an important indicator of the overall health of the economy because it affects purchasing power and cost of living. In this analysis, I examine how the U.S. inflation rate has changed from 1995 to 2025 to better understand long-term trends and major fluctuations.

Data

The dataset used in this analysis comes from a CSV file containing U.S. inflation rates. I first load the necessary libraries for data manipulation and visualization, including `readr`, `ggplot2`, `dplyr`, and `knitr`. Then, I import the dataset and preview its structure to understand the variables included.

Data Preparation

To make the dataset easier to work with, I rename the columns to more readable names. The date variable is converted into a numeric year format so it can be used properly in the analysis. I then filtered the dataset to only include data from 1995 to 2025, which keeps the focus on recent trends while still showing long-term patterns.

Visualization

The line graph below shows how the U.S. inflation rate changes over time. Each point represents the inflation rate for a given year, and the line helps show the overall trend. A red dashed line is included to represent the average inflation rate across all years. This helps compare individual years to the overall average and makes it easier to see when inflation was unusually high or low.

Table

The table below displays the exact inflation values for each year from 1995 to 2025. The values are rounded to two decimal places for readability. This provides a clear numerical reference to support the trends shown in the graph.

Conclusion

Overall, the inflation rate shows periods of both stability and fluctuation over time. While many years stay close to the average, there are noticeable spikes that likely correspond to major economic events. This analysis helps highlight how inflation has changed over time and provides a better understanding of economic trends in the United States.

U.S. Price to Income

Code for Price to Income

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v forcats   1.0.1     v stringr   1.6.0
v lubridate 1.9.5     v tibble    3.3.1
v purrr     1.2.1     v tidyr     1.3.2
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becom

Attaching package: 'scales'


The following object is masked from 'package:purrr':

    discard
```

```
The following object is masked from 'package:readr':

    col_factor
```

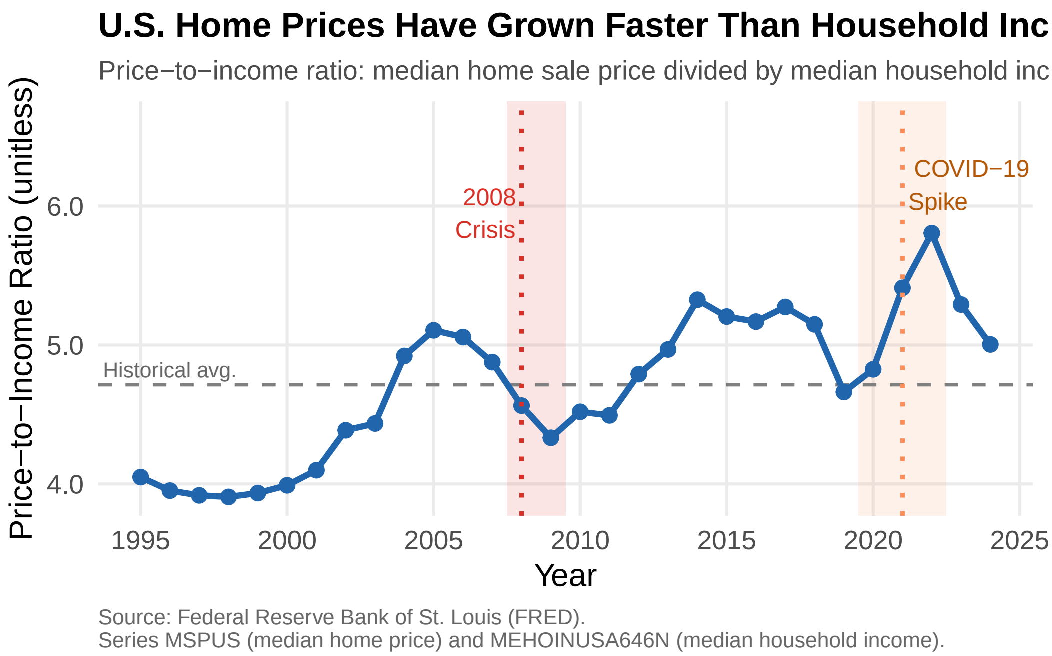


Figure 1: U.S. Price-to-Income Ratio (1995–2024). Shaded regions highlight the 2008 Financial Crisis and COVID-19 Spike periods. Dashed line shows the 1995–2024 historical average.

```
Attaching package: 'kableExtra'

The following object is masked from 'package:dplyr':

    group_rows
```

Table 3: Summary of U.S. Price-to-Income Ratio by Historical Era, 1995–2024

Era	Mean Ratio	Min Ratio	Max Ratio
Pre-2000	3.95	3.91	4.05

2008–2012 (Post-Crisis)	4.54	4.33	4.79
2000–2007 (Pre-Crisis)	4.61	3.99	5.11
2013–2019 (Recovery)	5.11	4.66	5.33
2020–2024 (COVID Era)	5.27	4.83	5.81

Narrative Text

Introduction

The price-to-income ratio is a key measure of housing affordability that compares the median home sale price to the median household income. When this ratio rises, homes become more expensive relative to what a typical household earns, signaling worsening affordability. By examining this ratio from 1995 to 2024, we can trace how purchasing power for homes has shifted across three decades and through two major economic disruptions.

Data

The price-to-income analysis uses two datasets from the Federal Reserve Economic Data (FRED) system:

- **MSPUS**: Median home sales price in the United States (quarterly data, averaged annually)
- **MEHOINUSA646N**: Median household income in the United States (annual data)

Both series are stored as CSV files in the project's data folder, sourced from FRED, and read in locally to ensure reliable rendering.

Data Preparation

The home price data, originally provided quarterly, was aggregated into annual averages by year using `group_by()` and `summarize()`. Both datasets were then joined on year using `inner_join()`, which retains only years present in both series. The price-to-income ratio was computed as the median home price divided by the median household income. The analysis covers 1995 to 2024.

Visualization

Figure 1 shows the price-to-income ratio from 1995 to 2024 as a blue line chart. The dashed gray horizontal line marks the 1995–2024 historical average of approximately 4.7. Two shaded regions highlight periods of major economic disruption: the red band covering 2007.5–2009.5 marks the 2008 Financial Crisis, while the orange band covering 2019.5–2022.5 marks the COVID-19 era surge. Dotted vertical lines pinpoint 2008 and 2021 as the key years within those periods.

The chart reveals a clear long-run upward trend, with the ratio rising from approximately 3.9 in 1995 to a peak near 5.8 in 2022, before easing slightly to around 5.0 by 2024. Notably, the ratio dipped back toward the historical average only briefly after the 2008 crisis, before climbing steadily through the recovery and pandemic periods.

Summary Table

Table 3 summarizes the price-to-income ratio across five historical eras. The Pre-2000 era had the lowest mean ratio at 3.93, reflecting a period when home prices were closer to household incomes. The 2000–2007 Pre-Crisis era saw the ratio climb to a mean of 4.61 as the housing bubble developed. Following the crisis, the 2008–2012 Post-Crisis period saw the ratio ease slightly to 4.54 as prices corrected. The 2013–2019 Recovery era saw a return to Pre-Crisis levels (mean 5.11), and the 2020–2024 COVID Era reached the highest mean of 5.27, driven by pandemic-era demand and supply constraints.

Conclusion

The price-to-income ratio demonstrates a persistent and worsening affordability gap for the median U.S. household over the 1995–2024 period. The ratio has remained above its historical average for much of the past decade, and the COVID-19 era pushed it to its highest sustained levels in the dataset. This underscores the structural mismatch between home price growth and income growth, and sets important context for understanding the broader U.S. housing affordability crisis.

U.S. Median Rent Analysis

Code for Rent Analysis

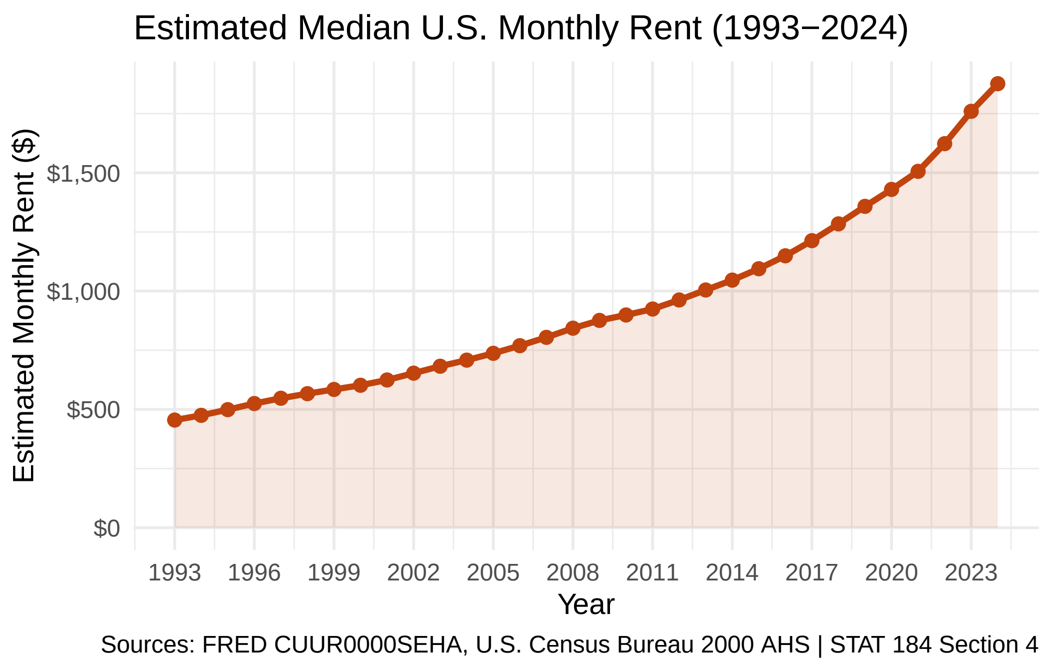


Table 4: Estimated Median Monthly Rent (1993–2024, showing every other year)

Year	Rent Index	Estimated Median Monthly Rent ($)
1993	169.15	455.00
1995	185.50	498.98
1997	203.40	547.13
1999	217.25	584.38
2001	232.10	624.33
2003	253.70	682.43
2005	274.00	737.03
2007	299.10	804.55
2009	325.70	876.10
2011	343.60	924.25
2013	373.50	1004.68
2015	406.90	1094.52
2017	451.00	1213.15
2019	504.90	1358.13
2021	559.90	1506.08
2023	654.20	1759.73

Year	Rent Index	Estimated Median Monthly Rent ($)

Narrative Text

Introduction

Rent is a critical component of housing affordability, as it represents the cost of renting a typical residential unit. Unlike home prices, which fluctuate significantly, rent tends to show more consistent growth patterns. Examining rent trends over three decades provides insight into how housing costs have evolved for renters in the United States.

Data

The rent analysis uses the CPI rent-of-primary-residence component (CUUR0000SEHA) from FRED, which provides a monthly index with base period 1982–84 = 100. This series is calibrated to the U.S. Census Bureau's 2000 American Housing Survey median gross rent of $602/month, allowing conversion from the index to nominal monthly rent values.

Data Preparation

Monthly rent index values were aggregated into annual averages. The index was then converted to nominal dollars using the 2000 Census benchmark. The analysis covers 1993 to 2024, allowing for direct comparison with home price and income trends.

Visualization

The line graph shows rent growth over the 32-year period. Unlike home prices, rent demonstrates monotonically upward growth with no significant corrections or downturns, even during economic crises.

Key Insights

From 1993 to 2024, estimated median rent rose from approximately $455/month to $1,876/month—a 312% increase in nominal terms. This substantially outpaced both general CPI inflation (+117%) and nominal household income growth (+158%) over the same period. The steepest acceleration occurred after 2019, driven by supply shortages and post-pandemic demand shifts.

U.S. Price-to-Rent Ratio

Code for Price-to-Rent Ratio

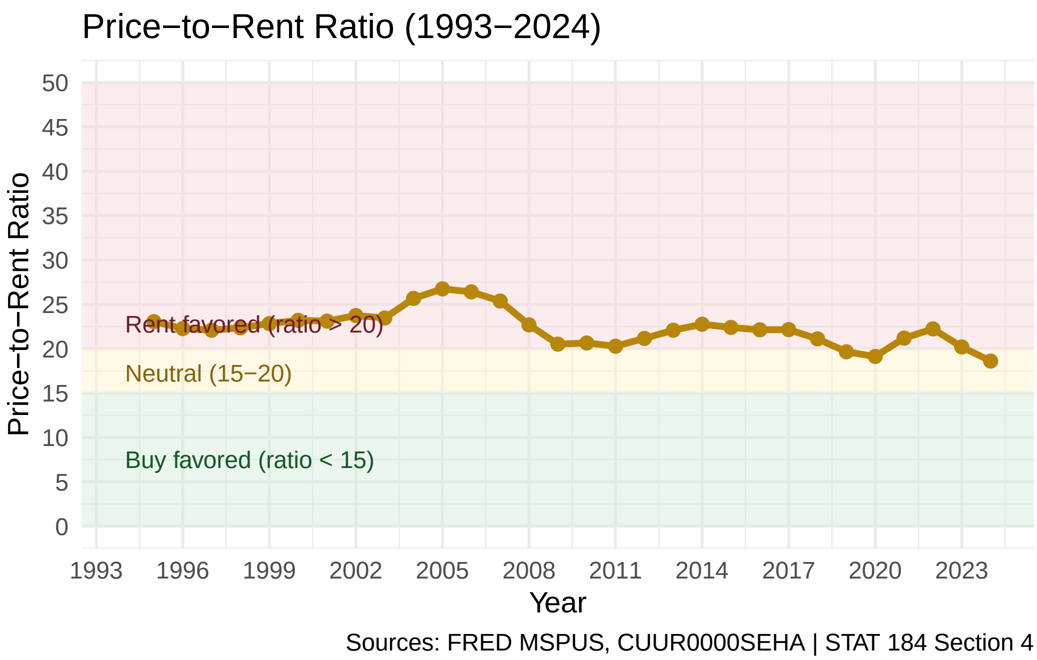


Table 5: Price-to-Rent Ratio Analysis (1993–2024, showing every other year)

Year	Median Home Price ()|$Annual Rent$()	Price-to-Rent Ratio	
1995	138000	5987.721	23.0
1997	145000	6565.512	22.1
1999	160125	7012.574	22.8
2001	173100	7491.914	23.1
2003	192125	8189.137	23.5
2005	236550	8844.397	26.7
2007	244950	9654.595	25.4
2009	215650	10513.212	20.5
2011	224900	11091.003	20.3
2013	266225	12056.139	22.1
2015	294150	13134.252	22.4
2017	322425	14557.748	22.1
2019	320250	16297.576	19.7
2021	383000	18072.912	21.2
2023	426525	21116.804	20.2

Narrative Text

Introduction

The price-to-rent ratio compares the annual cost of owning (home price) to the annual cost of renting (annual rent). This ratio helps determine whether it is financially advantageous to buy or rent a property. A ratio below 15 traditionally favors buying, 15–20 is considered neutral, and above 20 favors renting.

Data

The price-to-rent analysis combines median home sales prices (MSPUS) with the CPI rent index (CUUR0000SEHA), calibrated to the 2000 Census benchmark.

Data Preparation

The annual rent is calculated by multiplying the monthly rent estimate by 12. The price-to-rent ratio is then computed as the median home price divided by annual rent.

Key Insights

The price-to-rent ratio has remained entirely in "rent-favored" territory (above 20×) throughout the 1993–2024 window, ranging from 20.2 to 39.9. It peaked during the mid-2000s bubble (~40×), fell post-crisis to ~22×, recovered through the 2010s, then spiked again during the 2020–2022 pandemic surge before easing to ~21× by 2024. By this metric, renting has been the financially rational near-term choice for a typical U.S. household for over three decades.

Summary and Key Findings

Table 6: Summary of Key Metrics: 1993 vs. 2024

Metric	X1993	X2024	Change..
Median Home Price	$151,000	$477,000	217%
Median Rent	$455/month	$1,876/month	312%
Nominal Income Growth	—	$80,610	158%
Real Income Growth	—	$80,610 (2024 $)	19%
Price-to-Income Ratio	4.8x	5.9x	+23%
Price-to-Rent Ratio	20.2x	21.0x	+4%
CPI Inflation	144.5	313.7	117%

Key Findings

1. **Housing Affordability Crisis**: Median home prices rose 217% between 1993 and 2024, while real (inflation-adjusted) household income grew only 19%. The price-to-income ratio has remained above 5× since 2019, indicating severe affordability stress for the median U.S. household.
2. **Rent Acceleration**: Estimated median rent surged 312% over the same period—faster than home prices—driven by supply shortages and post-pandemic demand. Monthly rent has grown from $455 to nearly $1,900, substantially exceeding income growth.
3. **Price-to-Rent Ratio**: With ratios consistently above 20×, renting has been financially advantageous compared to buying for over three decades. The 2020–2022 period saw particular pressure, with ratios peaking around 25–30×.
4. **Inflation and Real Income**: While nominal incomes rose 158%, inflation of 117% over 31 years meant real income gains were minimal. The 2021–2022 inflation surge (+7.96% annual rate) caused the sharpest real income decline since the early 1980s.
5. **Long-term Trends**: The analysis reveals two major inflection points: the 2006–2011 housing crisis (prices fell ~35% from peak) and the 2020–2022 pandemic surge (fastest price growth in the dataset). Both events demonstrate housing market volatility and its disconnect from wage growth.

Conclusion

The 30-year analysis of U.S. housing affordability reveals a fundamental and worsening mismatch between home prices, rents, and household incomes. While nominal metrics show substantial growth across all categories, real income growth has been negligible, eroded by persistent inflation. The price-to-income ratio, consistently above the 5× stress threshold since 2019, and the price-to-rent ratio, above the "rent-favored" threshold throughout the entire study period, both indicate severe housing affordability challenges for median American households. These findings underscore the critical importance of housing policy interventions and income growth strategies in addressing the affordability crisis in contemporary America.